

Neuromorphic ASR

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- ▶ Similar to the Interspeech 2025 paper. <https://arxiv.org/pdf/2505.24721>
- ▶ Curiosity about how a transducer model, more involved than CTC, would behave on memristor.
- ▶ Study the impacts of offloading various components of the transducer model to device.
- ▶ Study the actual impact of noise
- ▶ Performance impact on Lexfree vs Tree searches.

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- ▶ Memristor programming (programming table from paper?) and non idealities
- ▶ Memristor crossbars and bitlines to represent model weights and thus the need of QAT.

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- ▶ Our matrix is 3 (model types) x 5 (search types) x 3 'stages' (baseline, qat, memristor)
- ▶ Primary tables could be 1 for each model type.
- ▶ Appendix - Tables across models, but same stages for easy comparison.
- ▶ Noise result that you mentioned earlier

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- ▶ Error analysis?

Reference